

SMARTBRIDGE — Data Flow Diagram (DFD Level 1)

System Data Movement for MedSafe AI

Date:	August 12, 2026	Team ID:	MEDSAFE_AI_01
Project Name:	MedSafe AI — Healthcare Safety & Preventive Awareness Platform	Max Marks:	2 Marks

DFD Element Specification for MedSafe AI

Symbol Type	Element Name	Description / Purpose
External Entity	User / Patient / Caregiver	Provides prescription images, drug query names, and symptom inputs.
Process 1.0	OCR & Text Extraction	Extracts text from images using PyTesseract and structures JSON using LLaMA 3.
Process 2.0	Interaction Checker	Matches drug names via RapidFuzz and checks against interaction rules.
Process 3.0	Symptom Triage Engine	Scans red-flag keywords, calculates 1-10 risk score, and formats local AI advice.
Data Store D1	Drug Interaction DB	Local SQLite/JSON database containing active drug severity pairs and warnings.
Data Store D2	Red-Flag Emergency Rules	Local dictionary of critical symptoms mapped to risk scores (1-10).

Visual Data Flow Mapping

[User / Patient] == (Uploads Prescription Image) ==> [Process 1.0: OCR & Text Extraction] [Process 1.0] == (Extracted Drug List JSON) ==> [Process 2.0: Interaction Checker] [User / Patient] == (Drug Query Pair) ==> [Process 2.0: Interaction Checker] [Process 2.0] == (Query Drug Names) ==> [Data Store D1: Drug Interaction DB] [Data Store D1] == (Interaction Severity Data) ==> [Process 2.0] [Process 2.0] == (Interaction Warnings & LLaMA Explanation) ==> [User / Patient] [User / Patient] == (Enters Symptoms) ==> [Process 3.0: Symptom Triage Engine] [Process 3.0] == (Match Symptoms) ==> [Data Store D2: Red-Flag Emergency Rules] [Data Store D2] == (Triage Weight & Emergency Risk Level) ==> [Process 3.0] [Process 3.0] == (1-10 Risk Score + Emergency Guidance) ==> [User / Patient]
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